Counterfactual Attribution in Cascaded Retrieval

Progress Report — Weeks 1–2

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Summary

The project asks which stage of a two-stage retrieval cascade is responsible for the quality of its output, and answers it by constructing explicit counterfactual worlds over a frozen document universe. Weeks 1 and 2 are complete. All infrastructure, all data artefacts and all pre-registered decisions are in place; no real Stage-2 scores have been computed yet, which is Week 3.

Concretely: the evaluation universe is frozen at configuration fingerprint `46d0807d8ed3` with 872,455 query–document pairs; the BM25 Stage-1 run reproduces published effectiveness exactly; the entire Stage-2 bill is one measured 27-minute GPU job; 415 tests pass; and every methodological decision that could bias a result is recorded in the design log with its reasoning, before the result exists.

Five defects were found and fixed in shared code during this period, four of them in checks that were silently vacuous — they passed while being unable to detect the failure they existed to catch. Those are described in Section 4, because they were the most valuable output of the fortnight.

1 The problem and the design

A cascade retrieves candidates $C_q = \mathrm{Top}_{K_c}(U_q, s_1)$ with a cheap Stage-1 scorer and reorders them with an expensive Stage-2 reranker s_2. Asking "which stage did the work" is not a question about score correlations; it is a question about what would have happened otherwise.

1.1 The degeneracy result, and why it motivates the design

For a *pointwise* Stage-2 scorer, once C_q is fixed the final ranking is invariant to s_1. Stage 1's only causal channel into the output is *membership* in C_q — not the scores it assigned. This is why score-level attribution cannot work here, and it is the result the whole experimental design is built around.

The complement matters as much: *under a budget constraint*, Stage-1 rank decides which candidates survive the cut, so it becomes causally active again. That is what the budget worlds measure.

1.2 Four worlds over a frozen universe

U_q is the union of both Stage-1 systems' top-M retrievals, frozen once. Every claim is relative to it, so regenerating it would invalidate every previously computed effect; this is enforced mechanically by a fingerprint sidecar rather than by convention.

World	Ranking	Question it answers
A (actual)	Sort(C_q, s_2)	what the cascade really did
B (no rerank)	C_q in Stage-1 order	what the gate alone achieves
C (no gate)	Sort(U_q, s_2)	what the reranker would do unrestricted
D (budget)	Sort(retained $\cup$ promoted, s_2)	what a realistic budget buys

1.3 Three estimands, which do not add up

$$E_{\text{rerank}} = U_d(\pi_{\text{act}}) - U_d(\pi_{\text{no2}})$$
$$E_{\text{gate}} = U_d(\pi_{\text{act}}) - U_d(\pi_{\text{no1}})$$
$$\Delta_t = U_d(\pi_t) - U_d(\pi_{t=1})$$

They are deliberately non-additive. U_d is the per-document contribution to DCG@k under an explicit gain map.

1.4 Conditional invariants: the central design point

The effects obey sign rules that hold only *conditionally on membership*: e_{rerank} is exactly zero off C_q; e_{gate} is ≥ 0 for survivors and ≤ 0 for non-survivors; and Δ_t has four membership strata with three of the four signs fixed (removed ≤ 0, selected outsider ≥ 0, unselected outsider $= 0$, retained either sign).

A single unconditional validator would therefore reject correct Δ_t rows while passing incorrect E_{gate} rows. Validation is stratified throughout, and `metrics.evaluate_within_strata` deliberately returns no pooled row: because sign(E_{gate}) is fixed by survivorship and three of four Δ_t signs by allocation membership, a pooled statistic would reward a model for detecting the partition rather than the effect.

1.5 The gain map

TREC-DL grade 1 means "Related" — on topic, does not answer the query. The frozen map is {0:0, 1:0, 2:3, 3:7}. Because every estimand is a *difference* of utilities, scoring grade 1 as relevant would not cancel; it would inflate every effect in which a merely related passage moves. Unmapped grades raise rather than defaulting to zero.

2 What was built

Component	Owner	Tests
`shared/` — engine, worlds, effects, strata, metrics, schema	all (reviewed)	224
`experiments_anupam/` — data, retrieval, universe, allocations	Anupam	145
`experiments_shubham/` — DPR, world runner, power, analysis	Shubham	46
`experiments_chaitanya/` — reranker, cache, features, predictors	Chaitanya	25 (in PR)
Total on `main`		**415**

Roughly 4,600 lines in `shared/` with 2,600 lines of tests against it, plus 2,800 and 2,100 lines in the Anupam and Shubham workstreams. Members never import each other's modules; all exchange happens through validated file schemas in `shared/schema.py`.

3 Results

3.1 Data and Stage-1 runs

TREC-DL 2019 and 2020 passage ranking: $43 + 54 = 97$ judged queries over MS MARCO, 20,646 judgments. Split by query 49/19/29 at seed 42, committed as a file so "should be identical" becomes "is identical".

BM25 (k_1=0.9, b=0.4, Pyserini prebuilt index) reproduces the published figures *exactly*:

	nDCG@10		Recall@1000	
	ours	published	ours	published
DL19	0.5058	0.5058	0.7501	0.7501
DL20	0.4796	0.4796	0.7863	0.7863

Delta 0.0000 on all four. This is the only check that verifies the run is *right* rather than merely well formed: a run built against the wrong index passes every schema and depth check. It is wired as a gate that exits non-zero.

The dense run is `castorini/cosdpr-distil`, top-5,000 for all 97 queries, contributed by Shubham.

3.2 The frozen universe

Fingerprint	`46d0807d8ed3`
Queries	97
$\lvert U_q \rvert$	7,045–9,986 (median 9,097)
Total pairs	**872,455**
Judged fraction of U_q	0.0165 (min 0.0026)

Reproduced bit-identical in three independent environments across two operating systems, which demonstrates most of the Week-4 reproducibility requirement early. The universe parquet itself is not committed: it is reproducible from the committed run tables plus the configuration, and the manifest records the fingerprint and size distribution so a rebuild is verifiable.

The Stage-2 bill is the universe, once. The cache keys on (query, doc) and every world re-ranks a subset of U_q, so the union across both Stage-1 systems, all three K_c, every t and every outsider policy *is* U_q. The entire project costs 872,455 cross-encoder calls — not that many per world. At Chaitanya's measured throughput this is a single 27-minute job, which is what made the reference-subset sampling design unnecessary.

3.3 Feasibility gate

Three of the four feasibility questions are settled favourably.

K_c	Jaccard	judged C_q (bm25/dpr)	density ratio	gain-carrying non-survivors
100	0.163	0.51 / 0.48	28.6× / 26.9×	6 / 6
500	0.136	0.178 / 0.158	10.9× / 9.3×	3 / 3
1000	0.115	0.108 / 0.094	6.4× / 5.7×	2 / 1

Leverage: confirmed. Candidate Jaccard of 0.11–0.16. Two Stage-1 systems selecting nearly the same documents would leave the cross-paradigm comparison nothing to separate, however different their scores.

Cross-paradigm confound: retired. We expected the TREC 2019/2020 pools to favour lexical runs, leaving BM25's candidate set better judged than DPR's. Measured, the ratio is 1.06×

at K_c=100 and 1.15× at 1000: those pools included neural runs. The floor is symmetric too — both systems retrieved exactly 5,000, so each floors $|U_q| - 5000$ and the floored fraction is an identical 0.450. The comparison is unconfounded on both axes that could have spoiled it.

Gate-failure supply: thin, and it drove a rule change. See Section 4.

Judgment placement: the one open validity threat. U_q is 1.65% judged while BM25's top-100 is 51% judged. World C sorts ∼9,000 documents of which almost none are judged, so judged survivors are displaced out of the top-10 there, $U_d(\pi_{\text{no1}}) \to 0$, and $E_{\text{gate}} \to \text{DCG}(\pi_{\text{act}})$ — the gate would appear to account for *all* retrieved utility as an artefact of where NIST placed judgments in 2019. Resolved procedurally at freeze v2; see Section 5.

3.4 Stage-1 over the universe, and the allocations

The outsider policies need Stage-1 beyond depth M: `boundary` reads s_1 below the cutoff and `alternative` reads the other retriever's ranks over $D_q = U_q \setminus C_q$. Each run was extended to all of U_q under the floor policy (an unretrieved document scores strictly below everything the retriever returned — exactly what it asserted, *not in my top-M*).

Policy supply turned out to be 40–49× the budget at every depth (boundary 4,900/4,500/4,000 genuinely ranked documents below the cutoff against budgets of 100/500/1000), so both policies are real rather than nominal: neither has to draw from the tied floored block where the selection would be the `doc_id` tie-break wearing a policy's name.

11,640 allocations were then published and audited — 1,319,200 promoted documents — with four invariants all holding: every allocation spends exactly B; the four membership strata partition U_q; t=1 reproduces the plain cascade, so every Δ_t is measured against something that actually ran; and the random control's selection changed under a reseed in 100% of settings, which is what makes it a control.

3.5 Reranker and power (contributed)

Chaitanya measured 535 pairs/sec on a Colab T4 (MiniLM-L-12-v2, `float32`), giving ∼27 minutes for the whole universe or ∼9 minutes each under a three-way shard plan. Mean sequence length ∼71 tokens, max ∼183, far inside `max_length`=512. The **shuffle-invariance test passed on the real reranker**, which verifies rather than assumes the pointwise property the entire E_{gate} derivation rests on.

Shubham's power simulation, re-run at the headline K_c=100 and the real median $|U_q|$ of 9,097, returns: no subset smaller than all 97 queries reaches 0.80 power, and even at 97 only the `alternative` outsider policy clears it (0.935 continuous, 0.875 group). `boundary` falls short (0.67, 0.41). Therefore **H7 is confirmatory for `alternative` and descriptive for `boundary`**, with `random` as the control, declared before any real effect exists.

4 Corrections and methodological findings

The most valuable output of the fortnight was five defects in shared code, four of them in checks that passed while being structurally unable to detect the failure they existed to catch.

1. The H7 power denominator was conditional. `required_reference_size` divided detections by the number of *testable* draws. A subset with fewer than two queries in a group cannot be tested at all, which is an H7 failure rather than a draw to discard, so the reported power answered "given the subset was testable" and overstated what a real subset delivers. Now unconditional, with the conditional figure available separately. The stale column was *removed* rather than kept, so a downstream rescaling that would have cancelled the fix raised an error instead and was found in the same change.

2. "Relevant" disagreed with the gain map, by 4–9×. `shared/qrels.py` counted `relevance > 0`, which includes grade 1, while the frozen map scores grade 1 as 0.0. A grade-1 non-survivor was therefore counted as a gate failure although excluding it costs no utility. Measured: at K_c=1000 the DPR candidate set shows 9 excluded relevant documents under the loose definition and exactly 1 that carries gain. The headline K_c was about to be chosen from the first number. Both functions now take the gain map as a *required* argument, so the definition travels with the configuration that set it.

3. `trec_eval` uses linear gain, not exponential. The BM25 verifier must match the definition that produced the published figures. Implemented from memory as $2^{\text{rel}} - 1$, it disagreed with `pytrec_eval` immediately (0.0799 against 0.1116 on the first random case). `ndcg_cut` uses gain = rel. The wrong form would have "verified" the run against a number it structurally cannot reproduce, sending us hunting a nonexistent index problem. Pinned by tests that cross-check against the real `trec_eval` C code where installed.

4. The headline-K_c rule was mis-specified (self-correction). The pre-declared rule — largest K_c with median gain-carrying non-survivors ≥ 1 — selected K_c=1000, where the median supply is 1–2 and 36–39 of 97 queries have none at all. The error was conflating two jobs: "≥ 1" is a *non-degeneracy floor*, and used as a *selection* rule it chooses the deepest depth, which is exactly backwards — a TREC pool is built from the top ∼100 of the systems of the day, so deeper K_c swallows more of the judged pool into C_q and leaves *less* supply outside it.

Separated, the rule takes the depth with greatest supply among those clearing the floor, which selects K_c=100 (median supply 6). This is a *post-hoc change to a pre-declared rule* and is recorded as one. It is defensible only because gate-failure supply is measurable *before* any effect is computed, so the choice cannot be driven by effect sizes; no threshold was tuned, and only the direction of the preference changed. A test feeds the rule a synthetic collection whose supply grows with depth, where it correctly selects a deeper headline — the check that it is a rule and not a constant in disguise.

5. The allocation spend check was vacuous. `validate_allocation` compared the submitted count against $\min(n_{\text{out}}, |\text{selected}|)$ — the amount *achieved* — so a short allocation always matched its own shortfall and passed. It accepted an allocation submitting 1 of a budget of 2 with an outsider still available in D_q: precisely the "one point of the t curve cheaper than the others" case its own docstring says it exists to refuse. The ceiling is the available pool, not the draw. Since `worlds.py` runs this on every world-D allocation, the stricter check now applies there too; nothing was relying on the loose behaviour, verified against all 11,640 real allocations.

6. Feature leakage in the predictors (open). Found by Shubham in review, confirmed and sharpened: the Stage-2 features are built for all top-5,000 documents rather than for C_q, while the code states "Extra Stage-2 calls = 0". For a document outside C_q a cached Stage-2 score exists *only because world C was computed.* The consequence is stronger than mis-stated cost: for a non-survivor d, $\text{rank}_{\pi_{\text{act}}}(d)$ is undefined, so $e_{\text{gate}}(d) = -U_d(\pi_{\text{no1}})$, and $U_d(\pi_{\text{no1}})$ is fixed by $\text{rank}_C(d)$, which the s_2 scores over U_q determine exactly. The feature does not approximate the target; it contains it. Note the 25 tests pass either way — the suite never asserts the C_q boundary for Stage-2 features, which is why it survived review-by-test. Stage-1 features over U_q remain legitimately free, since s_1 costs no GPU.

5 Freeze history

Freeze v1 (26 Oct), `46d0807d8ed3`**.** Dense Stage 1 is DPR (`cosdpr-distil`, replacing ANCE); Stage-1 systems and universe sources are BM25 and DPR; Stage 2 is a pointwise MiniLM cross-encoder; data is TREC-DL 2019+2020; the gain map gives grades 0 and 1 zero gain.

Freeze v2 (29 Oct), `46d0807d8ed3` **— unchanged.** Both live decisions looked like configuration changes and neither was one, so the frozen universe, its sidecar and every manifest written against v1 remain valid.

Precision stays `float32`**.** `float16` would take the fill from ∼27 to ∼13 minutes: a saving of 14 minutes, once, for the entire project. Against it, every estimand is a difference of *rank-based* utilities, and `float16` perturbs scores at $\sim 10^{-3}$, which flips near-ties in a 9,000-document ranking. How many, and whether they land where they matter, is unquantified — and quantifying it would cost more GPU time than the saving. The cost side was already measured, so the decision needed no further evidence.

Unjudged competitors: compute every effect both ways, always. Condensed-list evaluation (Sakai, *Alternatives to Bpref*, SIGIR 2007) drops unjudged documents before ranks are assigned, so a judged document's rank counts only judged documents above it. Three reasons for computing both:

- **It costs no GPU.** Condensing transforms *rankings*, not scores — same cache, same 872,455 calls, CPU only.
- **It is the only option that does not decide in the dark.** The measurement that would settle which mode is right — the judged fraction of world C's top-10 — does not exist until the cache is filled. Every other option commits now to a choice whose evidence arrives later. Pre-registration exists to fix the *procedure* in advance, not to guess the answer.
- **It needs no configuration field.** If both are always computed there is no choice to encode, so the fingerprint does not move.

The *primary* estimand remains the uncondensed one, because it is what a deployed policy would really do; the condensed figure is reported beside it as a robustness check. If the two disagree in sign or in which stratum dominates, that disagreement *is* the finding and gets its own treatment rather than a footnote. Declared before any effect exists.

Also resolved: cache seeding from existing scores dropped (∼3 minutes of a 27-minute bill, against interleaving two scoring runs); SPLADE out (a third Stage-1 system would require rebuilding the frozen universe, and the Jaccard already gives ample leverage); hashing out per written criteria; reference subset all 97 queries. No open decisions remain in the log.

6 Commit history

Commit	Date	Subject
Week 1 — setup, shared contracts, Stage-1 runs		
`75a7d7f`	26 Oct	Initial commit
`0e9e293`	27 Oct	Shared code: engine, worlds, effects, strata, metrics, schema
`b5a6b86`	27 Oct	*PR #1* DPR Stage-1 run, cosDPR-distil, top-5000 (H1)
`80d9e21`	27 Oct	World runner and budget runner (counterfactual engine)
`f9def56`	27 Oct	E1 synthetic validation and the H7 power simulation
`479eeee`	27 Oct	Record E1 and H7 results; fix the E1 gate check
`3940aa2`	27 Oct	Re-record E1 and H7 manifests from committed code
`bb729df`	27 Oct	*PR #2* counterfactual engine, E1, H7 power (H2)
`20dbf56`	28 Oct	Loaders, split, retrievers, Stage-1 runner, universe builder
Week 2 — data artefacts, feasibility, decisions		
`594e855`	28 Oct	**Fix:** make H7 power unconditional; drop stale rescaling
`839e1f0`	28 Oct	BM25 run (top-5000, 97 queries) and the frozen universe
`1b0df38`	28 Oct	Re-record the universe manifest (identical universe)

Commit	Date	Subject
`90cfc0d`	28 Oct	Week-1 handoff note; refresh stale README claims
`8a099e3`	29 Oct	Re-record H7 power against the fixed diagnosis module
`fb66cac`	29 Oct	Import torch lazily; skip DPR tests where torch is absent
`0adfed4`	29 Oct	H7 power at the feasibility report's headline K_c
`35155fa`	29 Oct	Effect analysis: distributions, labels, additivity, diagnosis
`4fb5f7d`	29 Oct	Restate H2 at headline K_c=100 and the frozen $\|U_q\|$
`36392db`	29 Oct	Record Week-2 decisions, throughput and H2
`427faf5`	29 Oct	Drop the `test_dpr` ignore from `make test-all`
`8f779a5`	29 Oct	*PR #4* effect analysis, H2 at K_c=100, lazy torch
`e1caac9`	29 Oct	Extend Stage-1 to U_q under the floor policy
`b6f8930`	29 Oct	Publish the reference subset and the t-allocations, with audits
`66b8904`	29 Oct	**Fix:** cap expected spend by what was available
`0f56ec9`	29 Oct	**Freeze v2**: fingerprint unchanged, all decisions recorded
`d3d5b9c`	29 Oct	Add the condensed ranking mode to every effect (freeze v2)

Twenty-nine commits across four merged pull requests. Shared-module changes go via pull request; each member's own directory goes direct to `main`. **PR #3** (Chaitanya, +3,076 lines across 12 files) is open pending the leakage fix.

7 Current state

Configuration fingerprint	`46d0807d8ed3` (v1 = v2)
Queries / judgments	97 / 20,646
Split (seed 42)	49 train / 19 val / 29 test, by query
BM25 verification	exact, $\Delta = 0.0000$ on all four measures
$\|U_q\|$	7,045–9,986 (median 9,097)
Universe pairs	872,455
Stage-1 floored fraction	0.450, identical for both systems
Headline K_c	100
Reference subset	all 97 queries
Allocations audited	11,640
Stage-2 bill	872,455 pairs, ∼27 min on one T4
Tests passing	415

8 Future work

8.1 Week 3 — fill the cache, compute real effects

1. **Blocking:** the PR #3 leakage fix, after which the synthetic fidelity numbers must be regenerated and all three baselines will move.
2. Fill the Stage-2 cache over U_q under the three-way shard plan: ∼9 minutes each, resumable and shardable with a merge step.
3. Compute E_{rerank}, E_{gate} and Δ_t over the real universe, in both the uncondensed and condensed modes, with stratified validation throughout.

4. **The decisive measurement:** the judged fraction of world C's top-10. If it is ∼20% unjudged, the judgment-placement concern is a footnote; if ∼95%, E_{gate} as primarily specified is not measuring the gate and the condensed figure becomes the headline.
5. Diagnosis labels and the H7 test, with group sizes reported beside every p-value, confirmatory only for `alternative`.

8.2 Week 4 — analysis and reporting

Effect distributions by stratum; the cost–fidelity curve, which must converge to the exact reference at full budget; utility versus t; the H7 comparison; a worked gating-failure example. Predictor fidelity against all three baselines (zero, membership-only, stratum-mean), fitted on training queries only. Full compute accounting. The reproducibility pack — already largely demonstrated by three bit-identical universe rebuilds across two operating systems.

8.3 Known risks

- **Judgment sparsity** is the main threat to E_{gate} and is mitigated procedurally rather than solved. The decisive measurement is scheduled for the moment scores exist.
- **H7 is descriptive for `boundary`** at any feasible subset size. This is a declared limitation, not a result to be improved by tuning.
- **Gate-failure supply is thin even at the headline depth** — a median of 6 gain-carrying non-survivors, with 17–23 of 97 queries having none. Per-query effect distributions will be noisy and must be reported with their supply.
- **Grade-1 prevalence** meant the usable relevant supply was 4–9× smaller than the naive count. Any future collection needs the same check before its K_c is chosen.

Reproducing this state

```
git clone https://github.com/a-n-u-p-a-m/IRE_PROJECT.git && cd IRE_PROJECT
pip install -r requirements.txt
make test-all                                        # 415 tests
make smoke-week1                                     # whole chain, no GPU, ~5s

python -m experiments_shubham.fetch_trec_dl          # queries + qrels
python -m experiments_anupam.data_prep               # qrels parquet + split
python -m experiments_anupam.run_stage1 --system bm25      # needs Pyserini
python -m experiments_anupam.verify_bm25             # gate: exact or exit 1
python -m experiments_anupam.build_universe --qrels data/qrels.parquet
python -m experiments_anupam.feasibility_report
python -m experiments_anupam.stage1_over_universe
python -m experiments_anupam.reference_subset
python -m experiments_anupam.allocations
```

Large derived artefacts are not committed; they are reproducible from the committed run tables plus the frozen configuration, and every manifest records the fingerprint it was produced under. Full reasoning for every decision above is in `docs/design_log.md`.